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Password Protected Image Encryption and Decryption using Hybrid Technique for Data Security

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Abstract

Data security and data anti-corruption are some areas of cryptography which is under continuous research and development. In this paper, a password based protection scheme has been included in order to increase the level of security of data. Here a hybrid technique for data security has been proposed and compared with its non-hybrid counterparts which resulted in the hybrid technique giving better results for various parameters of study like the mean square error and the signal to noise ratio. The proposed Hybrid Technique uses the Discrete Wavelet Transform (DWT) watermarking technique and Secure Force (SF) algorithm for cryptography. In this paper, only image data has been considered and the simulations have been carried out using MATLAB.

Keywords: *Cryptography (SF-Algorithm), DWT Watermarking, Entropy, MSE, Password, PSNR, SNR*

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INTRODUCTION

Connectivity using the cloud has gained popularity in the field of data communication. This feature though having its own pros is saddled with the major issue of data theft and hacking. Creating a strong and secure data safe and Safeguarding against data breach had become a major area of research in the area of Cryptography. Security of data is no longer a luxury but has become a necessity. Developing new encryption is a way to handle these situations.

Cryptography

The word 'Cryptography' comes from 'Kryptos' which meaning is 'hidden secret'. Cryptography, as the name suggests, is any method by which some secret message, hidden from unwanted hackers communicated to the legitimate user so as to reduce or remove the risks of data theft. This secret message can be converted into a format intended for the legitimate receiver's use. The cryptography technique is classified into two, namely symmetric/ privet key [1] and asymmetric/ public key. These secure applications are heavily dependent on various cryptography services, namely confidentiality, authentication and data integrity. Based on these cryptography services, the cryptography techniques and protocol are classified into four main regions. Which are; symmetric encryption, asymmetric encryption, data integrity techniques, and authentication protocols. Cryptography finds a wide range of applications in the domain of e-transactions, data broadcasting, etc. [2].

Security attacks are a threat to data safety. There are two types of security attacks, namely passive attack and active attack. In passive attacks, the attackers attempt to find out or use information from the system, without affecting the system resources. But in an active attack, an attacker tries to introduce unwanted data into the system as well as potentially change or modify the system resources or affect their operation.

Secure Force (SF) Algorithm

The SF Algorithm offers very low complexity architecture in Wireless Sensor Network (WSN) implementation. This is a four-step algorithm, namely (i) Key Expansion, (ii) Key

Management Protocol, (iii) Encryption Process and (iv) Decryption Process based on simple mathematics operation that operates on a four-bit data [3].

If we start with a 64-bit linear array data, it is first decomposed into 2 equal parts of 32 bit. Then again it is subdivided each 32-bit data into 4 equal parts of 16-bit data. Then all these parts are operated on row-wise Left Shift (LS) operation. This process is again operated on column-wise. Here XOR, XNOR, AND, and OR logical operations are implemented. The output data which is 64-bit data passed through the P-table using permutation and T-table using transpose. The output data is multiplied with fix matrix. Here we are using Lightweight Extensible Authentication Protocol (LEAP) as the key management protocol. Here all five keys which are K1, K2, K3, K4 and K5 created and recognized by using secure communication channel between the two points of contact of the sensor network. The encryption and decryption process is almost the same in this algorithm. Figure 1 shows the process of SF algorithm.

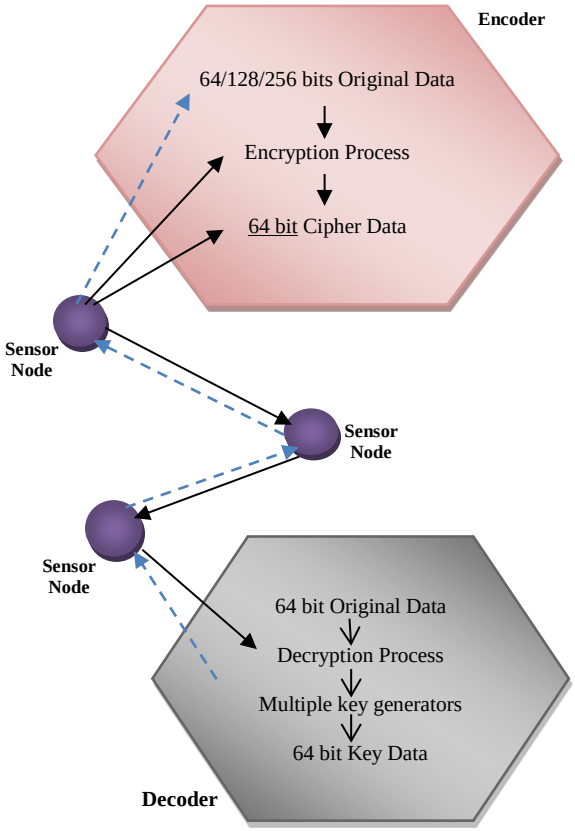

Fig. 1: *Flow Diagram of SF Algorithm.*

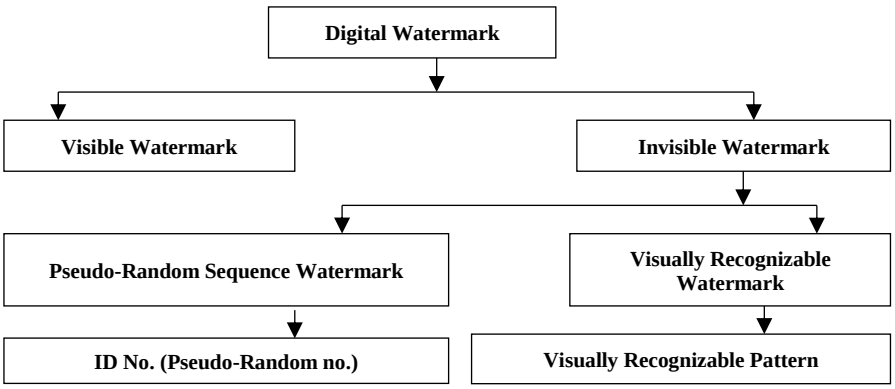

Fig. 2: *Types of Watermarking.*

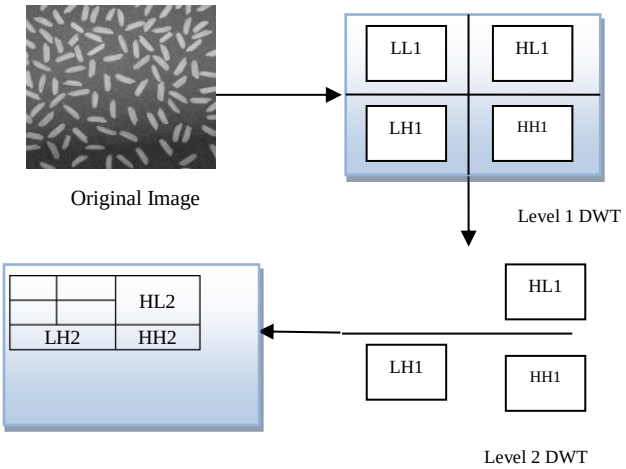

Fig. 3: *Level Dwt Watermarking Technique.*

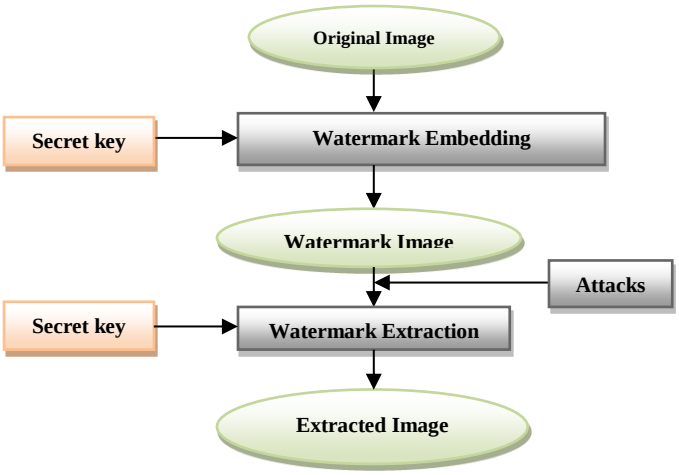

Fig.4: *Hows The Procedure of Watermarking Technique.*

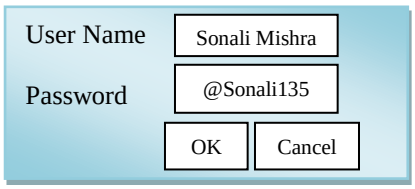

Fig. 5: *Simple Login Password Design.*

Password

A password is a word or string of characters used for user authentication to approve access to secured resources.

Nowadays, passwords are used for controlling user access to computer operating systems or mobile phones for protection. A set of alphabet, numerals and special characters is used in a combination to frame a phrase which can be used to access the resource in a secured manner. These phrases have restriction such as case-sensitivity and 8 characters minimum length for enhanced security purpose. Hence, the password can be perceived as a word that is used to pass through a secured system for accessing resources. For example, @sonali135 is a password of 8 characters length with lower- and upper-case alphabets, numbers and symbols. A password has to be short for the purpose of easy typing and remembering. Password policies are specified by many organizations by placing specific constraints such as length, alphabet case, numbers and special characters, etc. Figure 5 shows the sample login password design.

There are four levels at which a system can be attacked- Physical, Human, Operating system and Network [7].

PROPOSED WORK

In this paper a hybrid technique of encryption is proposed which is also compared with a

non–hybrid encryption technique to study the effect of encryption on data quality. Figure 6 shows the proposed hybrid technique.

The proposed Hybrid technique includes Watermarking, SF Algorithm Cryptography and Password. The password 1 is currently set as a known random number (known to both sender and receiver). The Password 2 may be modeled as any fixed identification number of the receiver (date of birth of the receiver is considered). We have considered some standard MATLAB images as our data and have analyzed the effect encrypted data quality based on various parameters as seen below. Here we have considered different images with a fixed dimension. When transmitted over a communication channel these data may be corrupted by noise. This has been modeled in MATLAB by considering Gaussian Noise.
Mathematically:

$$MSE = sum(sum(input\ image - noise\ image).^2)/(M * N) \quad (1)$$

Where; M and N- Dimensions of Images

$$SNR = 20 * log10((max - \text{mini})./\text{MSE} \quad (2)$$

$$PSNR = 10 * log10((256 * 256)./MSE) \quad (3)$$

$$Entropy = -sum(p.* \log_2 p) \quad (4)$$

Where, p is the normalized histogram counts which are returned from the MATLAB inbuilt function- *imhist*.

RESULTS AND SIMULATION

We have to use MATLAB for our simulation work. First, the various image data were transmitted with SF algorithm encryption only. Various parameters like the PSNR, MSE and SNR of the different data sets were calculated and recorded. Figure 7 shows the output of different received images which were transmitted with SF algorithm encryption.

Figure 8 shows the original, watermark embedded, SF encrypted, password box to extract the de-watermarked image, de-watermarked image, password box to extract the decrypted image and decrypted images of Lena by using hybrid technique. For the Watermarking we have used the DWT technique. Figure 9 and Figure 10 show the original, watermark embedded, SF encrypted, de-watermarked image and decrypted images of Microsoft Logo and Baboon respectively by using the hybrid technique.

The different parameter like PSNR, MSE and SNR of the different decrypted images were calculated and tabulated.

Table 1 shows the calculation of the same for different images using SF algorithm (non-hybrid encryption technique).

Different images were taken to increase the reliability of our study. The different values tabulated above are for SF decrypted images only.

In Figure 8 we start with the original data to which Watermark is added using DWT decomposition technique to get the watermarked image Figure 8 c and then SF algorithm is applied to encrypt the image data as can be seen in figure 8 d. At the receiver, end passwords are used for enhanced security while the first password gives the de watermarked image the second password gives us the decrypted image as seen in figure 8 h. The same process was applied to all the other data and the outputs can be seen in the Figure 9–10.

Likewise, the implementation of the hybrid technology on the Microsoft logo and baboon standard MATLAB data can be seen in Figure 9 and 10.

Table 1: *Parameter Calculation Using SF Encryption.*

Parameters	Lena	Microsoft Logo	Baboon
MSE	75.56	70.78	74.40
PSNR	29.38	29.67	29.45
SNR	10.57	11.13	10.70

Table 2: *Parameter Calculation using Hybrid Technique.*

Data→ Parameters	Lena	Microsoft Logo	Baboon
MSE	60.35	55.36	67.47
PSNR	30.45	30.73	29.87
SNR	12.52	12.92	11.55

Table 2 shows the calculation of MSE, PSNR and SNR of different images using a hybrid technique. A comparative study of different parameter calculated for each case follows.

Figure 11 shows the MSE values of different images and compares them between only using SF algorithm and using the Hybrid algorithm. From this graph, we can see we get better result of using a hybrid algorithm than using SF algorithm on MSE.

Figure 12 shows the graph of PSNR of different images and compares these images between using the hybrid algorithm and only using SF algorithm. From this graph, we can see that the PSNR value of different images using a hybrid algorithm gets the best output than using SF algorithm.

Figure 13 shows the graph of SNR values of different images and compares those images between using a hybrid algorithm and only using SF algorithm. From this graph, we can see that we get better result of SNR of using hybrid technique than only using SF algorithm.

The hybrid technique implemented in this paper was compared with some previous work as can be seen from the tabulations below.

Table 3: *Parameter Calculation of SF Algorithm with Affine Transform, SF Algorithm and Hybrid Algorithm.*

Techniques	MSE	PSNR
SF Algorithm with Affine Transform [8]	541.49	10.7949
SF Algorithm	75.56	29.38
Hybrid Algorithm	60.35	30.45

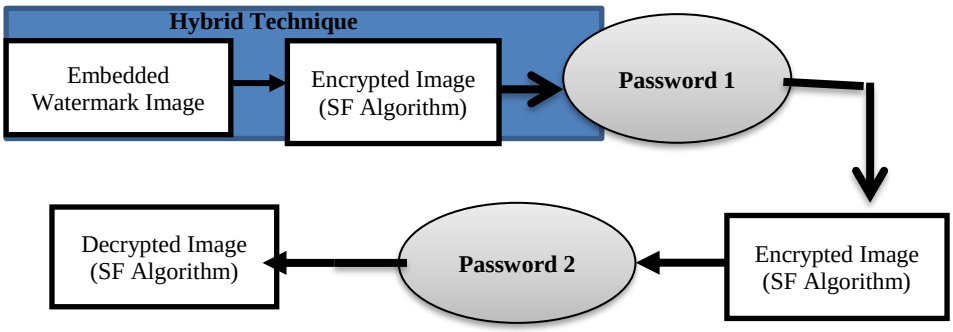

Fig. 6: *Proposed Hybrid Technique.*

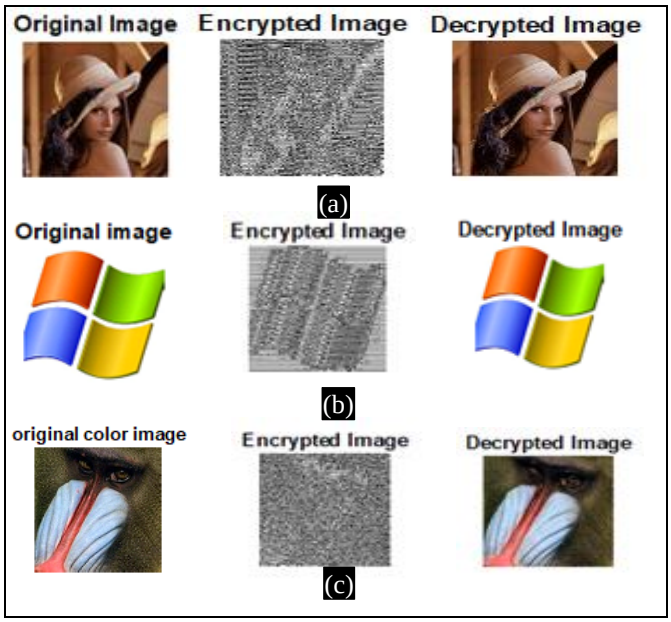

Fig. 7: *Reception of Different Transmitted Image Data Using SF Algorithm (a): Lena (b): Microsoft Logo (c): Baboon*

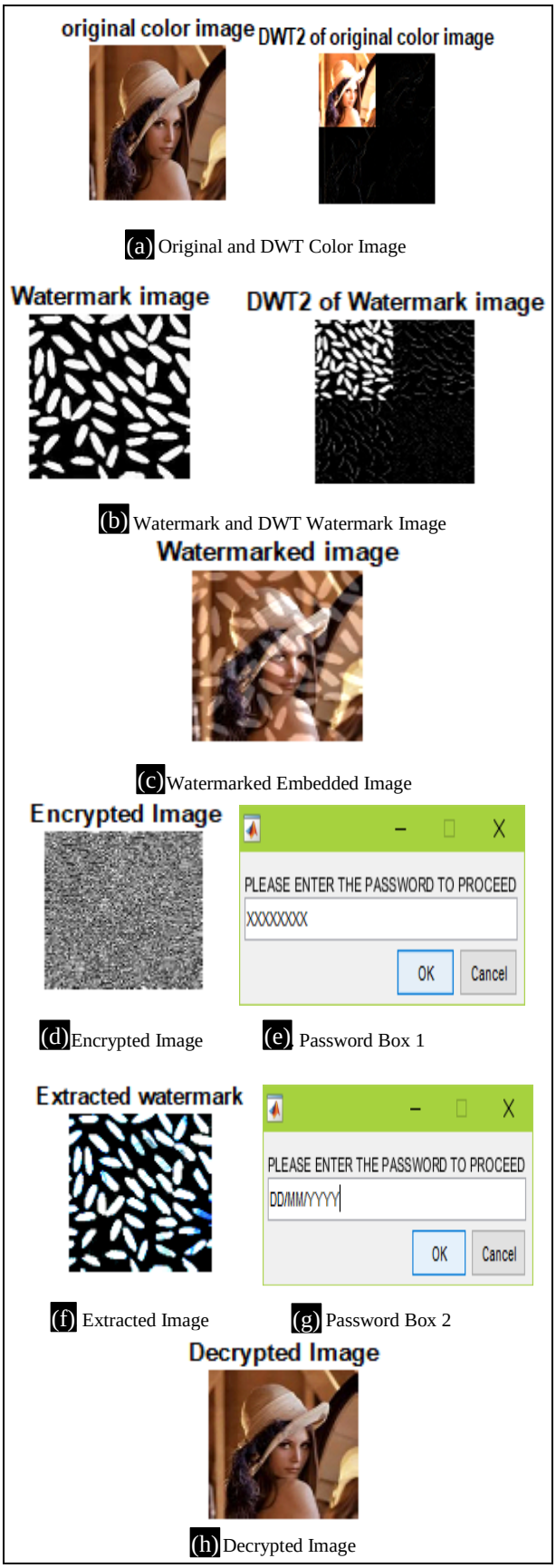

Fig. 8a-h: *Hybrid Encryption and Decryption Process.*

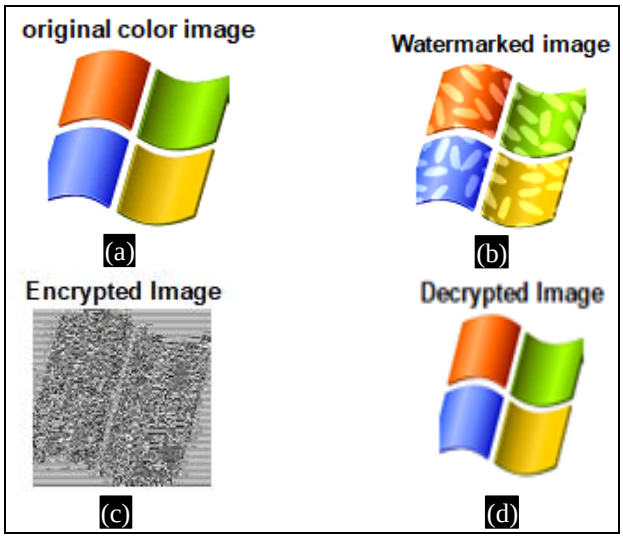

Fig. 9: *Reception of Microsoft Logo Image Data with Hybrid Encryption and Decryption a: Original Image b: Watermarked Image c: Encrypted Image d: Decrypted Image*

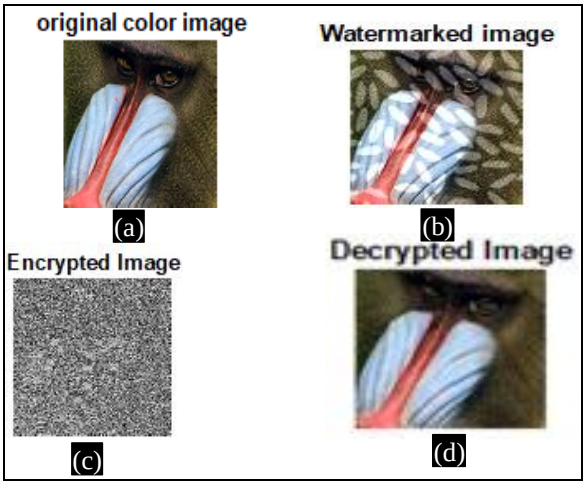

Fig. 10: *Reception of Baboon Image Data with Hybrid Encryption and Decryption; (a): Original Image (b) Watermarked Image (c) Encrypted Image (d) Decrypted Image.*

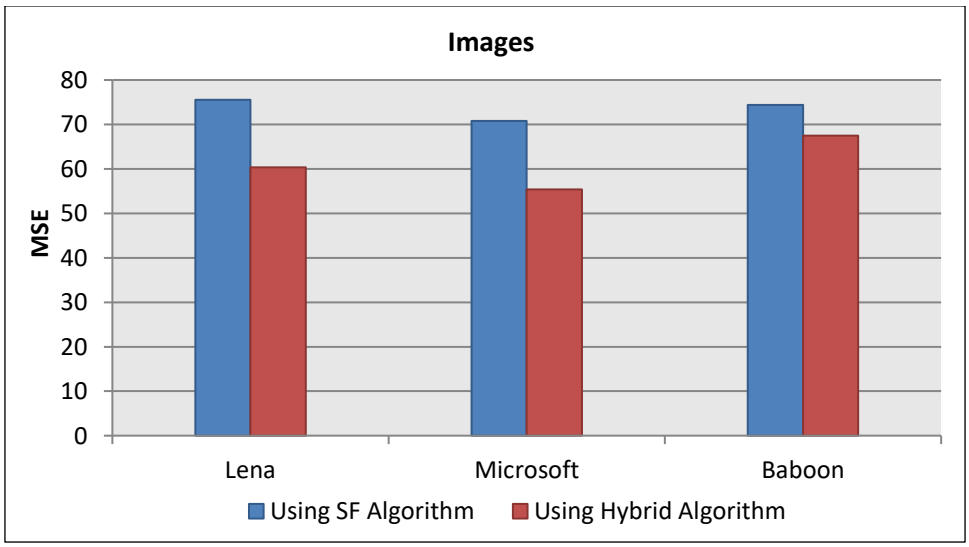

Fig. 11: *Graph of MSE Values of Different Images.*

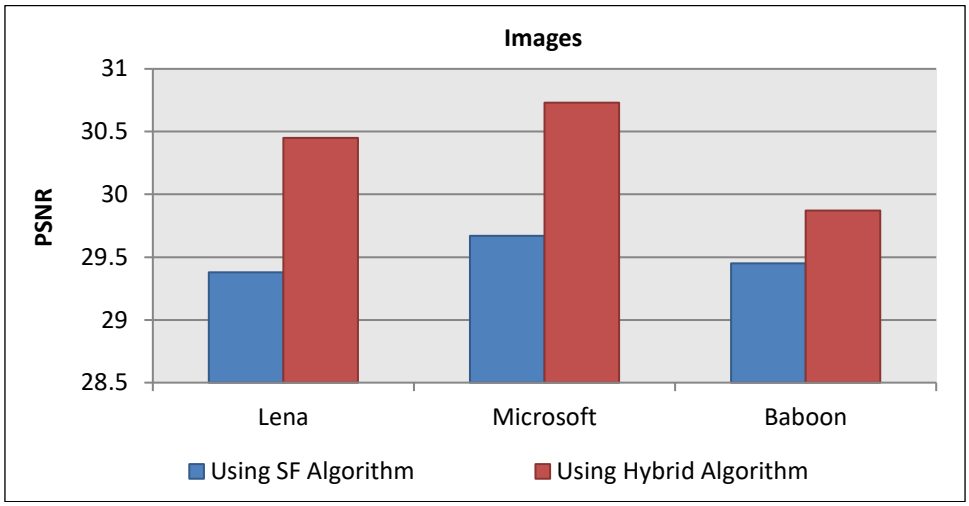

Fig. 12: *Graph of PSNR Values of Different Images.*

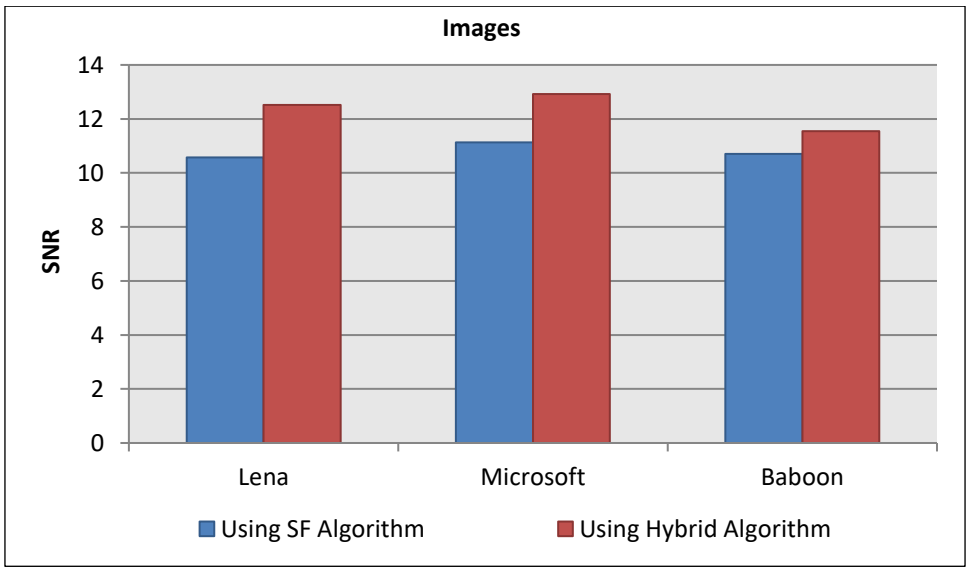

Fig. 13: *Graph of SNR Values of Different Images.*

From the Table 3, a comparison of MSE and PSNR value of Lena image using SF algorithm with affine transform [8], only SF algorithm and the hybrid algorithm has been done. From these comparisons, we can see the MSE and PSNR value of the hybrid algorithm is giving better results than the other algorithm techniques. Hence we can say data security can be enhanced by using the proposed hybrid technique of encryption.

CONCLUSION

In this paper, a hybrid encryption technique using SF algorithm and watermarking was proposed. Different parameters like MSE, PSNR and SNR of different images using hybrid technique and non- hybrid technique was calculated and compared. The comparison shows that for all kind of data coloured or black and white the Hybrid technique gave promising results for MSE, PSRN and SNR. Our work was compared with a previous work implementing a hybrid algorithm using SF algorithm with affine transform and we saw that our technique was giving better results with respect to different parameters of the study. Other algorithms can be studied for a hybrid technique to optimize the results further.

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